B.E. Fifth Semester (Information Technology) (CBCS) Winter 2022

Database Management Systems - 5 IT 01

P. Pages : 2 **AC - 3058**

Time : Three Hours * 1 6 0 5 * Max. Marks : 80

Notes : 1. Due credit will be given to neatness and adequate dimensions.
2. Assume suitable data wherever necessary.
3. Retain the construction lines.
4. Illustrate your answer necessary with the help of neat sketches.
5. Use of pen Blue/Black ink/refill only for writing the answer book.

SECTION – A

1. a) Explain different types of data models in detail. **7**

b) Explain the drawbacks of traditional file approach. How these are eliminated in database approach. **7**

OR

2. a) Explain following terms: **7**
i) Mapping constraints
ii) Multivalued attributes
iii) Derived attributes

b) Describe the architecture of database system and explain each component & it's types in detail. **7**

3. a) Explain following terms: **6**
i) Functional dependency ii) Domain & data dependency

b) Explain join operation & integrity constraints and it's type of relational data model. **7**

OR

4. a) Explain the following terms. **6**
i) Relational calculus ii) Relational algebra

b) What is Normalization? Explain 1NF, 2NF and 3NF with example. **7**

5. a) What is SQL? Explain data types, operators, wildcards and expressions. **6**

b) Explain SQL Aggregate functions count, sum, average, max, min & date function. **7**

OR

6. a) Explain following SQL clauses. **6**
i) Having
ii) Group by
iii) Order by

b) Explain following SQL commands with example and it's syntax. **7**
   i) Select and show database
   ii) Alter command
   iii) Insert command
   iv) Update command

SECTION – B

7. a) What is transaction states? Explain different types of transaction states. **7**

b) How atomicity and durability of transaction is implemented by the recovery management. **6**

OR

8. a) Explain ACID properties of transactions. Explain the usefulness of each property. **7**

b) What is transaction processing? Explain different operations on transaction. **6**

9. a) Explain timestamp based protocols with it's advantages and disadvantages. **7**

b) What is serializability? Explain conflict and view serializability. **6**

OR

10. a) Explain three phases for validation based protocol. **6**

b) Explain multiple granularity in locking. **7**

11. a) What is SQL injection? Explain types of SQL injections with example. **7**

b) What is authorization? Explain access control, DAC and mandatory access control. **7**

OR

12. a) Explain intrusion detection system with it's method and compare with fire walls. **7**

b) What is authentication process? Explain role based access control models. **7**

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